

EEE 312: Digital Signal Processing I Laboratory

Project Proposal

Voice-Based Meal Verification System for Hall Dining

**Submitted By:
Group 7 (Section C1)**

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Date of Submission:
July 24, 2026

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I. Summary

This project proposes an upgraded, touchless acoustic access and verification system for residential hall dining facilities at BUET. Building upon the foundational logic of a previous senior group’s ”Hall Entry System”, we intend to radically modify their baseline to suit the uniquely challenging and non-stationary acoustic environment of a busy dining hall during meal hours. The proposed system will incorporate critical DSP and administrative upgrades—such as Rational Resampling for computational speed, Sub-200 Hz Replay Attack detection for security, Spectral Subtraction for babble noise suppression, and automated CSV database logging—resulting in a highly accurate, spoof-resistant, and context-aware dining access kiosk simulated entirely in MATLAB.

II. Introduction and Theoretical Background

To successfully engineer a voice verification system that works in a real-world dining hall, it is crucial to move beyond simple time-domain heuristics. Our project adapts the baseline codebase of a previous BUET ”Hall Entry” project. However, the previous project relied on full 44.1 kHz processing, static high-pass filtering, and simple energy thresholding—methodologies that fail completely when exposed to cafeteria babble and dropped dining trays. To solve these issues, our project implements the following core DSP theories taught in EEE 311:

II.1 A. Rational Resampling (Downsampling for Efficiency)

Smartphones capture audio at 44,100 samples per second. Processing full 44.1 kHz arrays during 1-to-N database matching creates massive computational lag. Since human speech formants lie well below 4 kHz, the Nyquist-Shannon Sampling Theorem dictates that $F_s = 8$ kHz is sufficient. Because $44.1/8$ yields a non-integer, we cannot simply use naive decimation. We will implement **Polyphase Rational Resampling** (interpolating by $L = 80$, filtering, and decimating by $M = 441$) to mathematically safely downsample the signal, speeding up matrix processing by over $5\times$.

II.2 B. Noise Suppression via Spectral Subtraction

A traditional static filter (like the $1 - 0.97z^{-1}$ pre-emphasis used by the senior group) cannot remove dining hall chatter because background voices share the exact same frequency band as the target speaker (300 – 3400 Hz). Instead, we utilize **Spectral Subtraction**. By capturing 0.5 seconds of silence prior to the student speaking, we estimate the ambient noise magnitude spectrum $|N[k]|$ via the Discrete Fourier Transform (DFT). We then subtract this noise ”recipe” from the noisy speech spectrum $|Y[k]|$:

$$|\hat{X}[k]| = \max(|Y[k]| - \alpha|N[k]|, \beta|Y[k]|) \quad (1)$$

The parameter β acts as a spectral floor to prevent artifacts known as ”musical noise”.

II.3 C. Smart Endpoint Detection (STE + ZCR)

The previous baseline project used a single-threshold energy check ($\text{Energy} > 0.1 \times E_{\max}$) to find where the speech starts. In a dining hall, a dropped metal tray creates a massive energy spike, falsely triggering the system. We solve this by implementing a dual-check **Voice Activity Detector (VAD)** combining Short-Time Energy (STE) with the Zero-Crossing Rate (ZCR). While a tray crash has high STE, its chaotic ZCR forces the system to reject it. A frame is only kept if:

$$(\text{STE} > \text{Thresh}_{\text{STE}}) \quad \text{AND} \quad (\text{ZCR}_{\min} \leq \text{ZCR} \leq \text{ZCR}_{\max}) \quad (2)$$

II.4 D. Speed-Invariant Matching (Dynamic Time Warping)

Standard time-domain cross-correlation (‘xcorr’) fails because speech phases vary wildly, and students speak at different speeds, causing temporal mismatch. We extract the DFT magnitude of 30 ms frames and use **Dynamic Time Warping (DTW)** to non-linearly warp the time axis, aligning the spectral formants of the test sequence X and the template Y by minimizing the distance metric $D(i, j)$ using dynamic programming:

$$D(i, j) = |X_i - Y_j| + \min\{D(i-1, j), D(i, j-1), D(i-1, j-1)\} \quad (3)$$

III. Needs and Problems Addressed

The current dining hall access protocol at BUET relies heavily on manual, paper-based meal cards. This creates several logistical problems:

- **Hygiene and Efficiency:** Physical card handling slows queues and transmits pathogens.
- **Resource Misallocation:** Paper cards are easily faked or shared with non-residents.

Critical Weaknesses in the Senior Code Baseline: While a previous group developed a general “Hall Entry System”, there has never been a dedicated project for Hall Dining Entry. We are taking their Hall Entry codebase as our baseline, but it has severe flaws making it unusable for a dining environment:

1. **Massive Computational Overhead:** Seniors ran 26-band MFCCs and $O(N \cdot M)$ DTW matrices on full 44.1 kHz signals. Running a 1-to-N loop across hundreds of resident students at 44.1 kHz causes perceptible kiosk lag.
2. **Failure under Cafeteria Babble Noise:** Seniors relied only on a static pre-emphasis filter ($1 - 0.97z^{-1}$) and frequency truncation (300 – 3700 Hz). Background chatter (other students talking in the hall) sits in the exact same frequency band, rendering pre-emphasis useless against babble noise.
3. **Fragile Single-Threshold Energy VAD:** Their endpoint detector only checked $\text{Energy} > 0.1 \times \max(\text{Energy})$. In a dining hall, a dropped plate or metal tray creates a volume spike that tricks the detector. Conversely, soft consonant sounds (‘S’, ‘F’, ‘T’) get accidentally chopped off.
4. **Volume Inconsistency:** No volume gain normalization was used. If a student stood 10 cm closer to the microphone during authentication than during registration, the DTW energy coefficients mismatched, triggering a False Rejection (FRR).
5. **Lack of Dining-Specific Business Logic:** Gate entry logic does not prevent meal double-dipping, manage meal session times (Breakfast/Lunch/Dinner), or track meal budgets.

IV. System Workflow Diagram

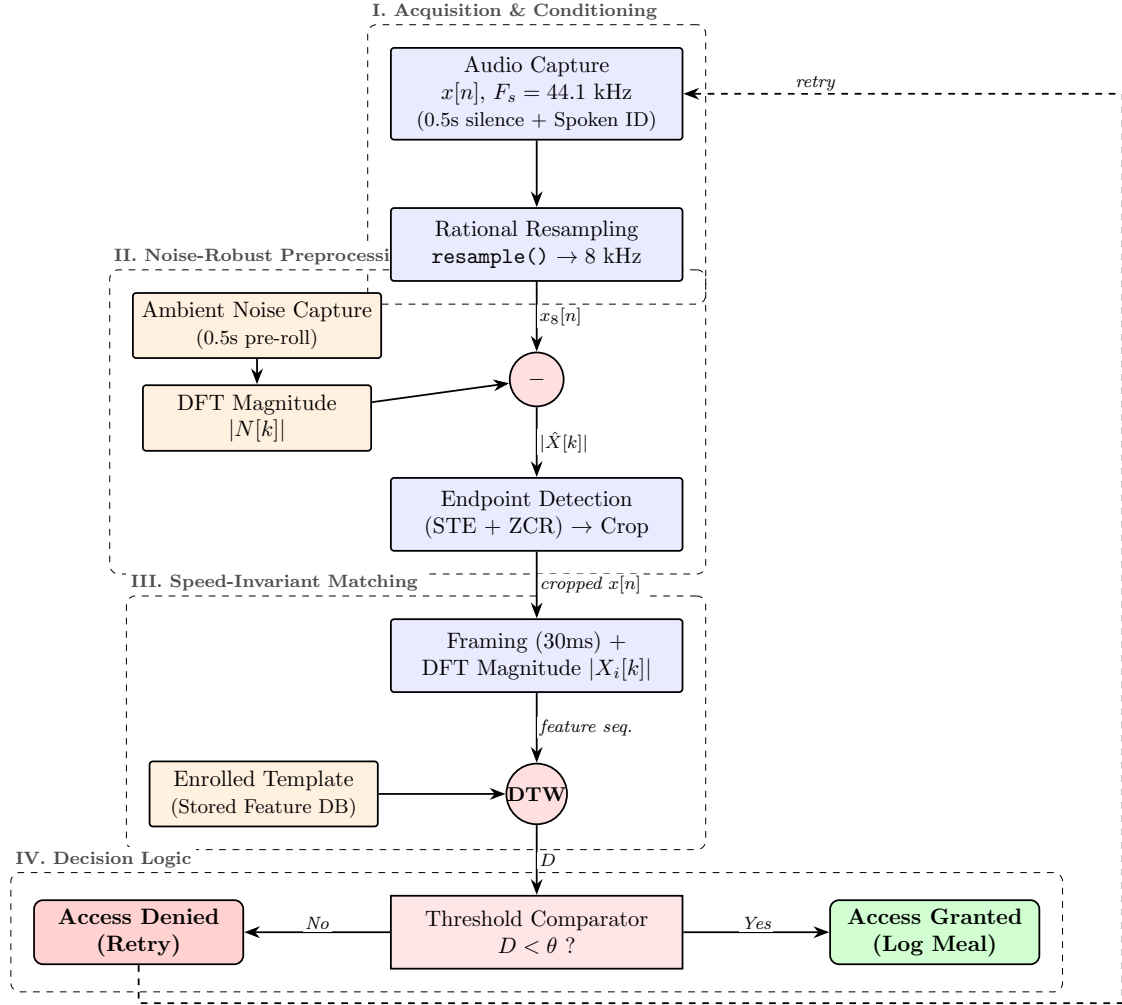


Figure 1: Signal-flow block diagram of the spoken-ID verification pipeline.

V. Goals and Objectives

This project aims to radically modify and upgrade the prior senior codebase to fulfill the EEE 312 COs:

- **Goal 1 (CO1 [PO(a), PO(e)]):** Implement classical DSP algorithms in MATLAB, including DFT (`fft`), Polyphase Resampling, and Dynamic Time Warping (`dtw`).
- **Goal 2 (CO2 [PO(d)]):** Investigate and compare the failure of strict time-domain correlation against the success of DTW-aligned spectral magnitude matching.
- **Goal 3 (CO3 [PO(c), PO(i)]):** Design optimal built-in anti-aliasing filters and a dynamic Spectral Subtraction algorithm to suppress background chatter.
- **Goal 4 (CO4 [PO(c), PO(l)]):** Enhance public health (zero-contact pathogen reduction) and environmental sustainability (eliminating paper waste).
- **Goal 5 (CO5 [PO(h)]):** Address residential ethics by utilizing acoustic matching to prevent proxy meals.

VI. Procedures and Modifications (Scope of Work)

We couldn't find any project regarding hall dining entry system. But we got a project report on **Voice Based Secured Residential Hall Entrance System by Group 1, EEE'21(A2)**. Taking it as baseline, we will implement the following distinct modifications over the baseline senior code:

1. **Audio Normalization & Resampling (Exp. 1 & 2):** The senior group processed un-normalized 44.1 kHz audio, which caused volume inconsistency errors and massive computational lag during matching. We will improve this by adding an Automatic Gain Control (AGC) step via RMS scaling to fix volume mismatches, and safely downsample the audio from 44.1 kHz to 8 kHz using polyphase resample to eliminate the kiosk processing lag.
2. **Sub-200 Hz Liveness Detection (Exp. 4):** The senior gate-entry project lacked any anti-spoofing security. We will improve this by introducing an acoustic liveness check. By analyzing the DFT energy ratio in the 0 – 200 Hz band, the system will detect and reject fake audio played from smartphone speakers.
3. **Noise Profiling and Spectral Subtraction (Exp. 4):** The seniors relied on a static pre-emphasis filter that failed to remove background babble noise because it shares the same frequency band as speech. We will improve this by discarding the ineffective filter and implementing dynamic Spectral Subtraction using a 0.5s pre-roll ambient profile to actively cancel cafeteria babble.
4. **Robust Endpoint Detection (Exp. 3):** The previous endpoint detector relied on a fragile single-energy threshold that easily triggers on dropped plates or chops off soft consonants. We will improve this by replacing it with a Hybrid STE + ZCR logic gate to ensure only true speech is processed.
5. **Dining Hall Business Logic & CSV Logging Implementation:** Because no previous group has built a Dining Entry system, the baseline code lacks any meal-specific logic. We will build an automated CSV database logging system utilizing `datetime('now')` to track meal attendance. This system will evaluate the current timestamp to verify if the kiosk is operating within permitted meal windows (Breakfast/Lunch/Dinner) and utilize a lookup function on the CSV log to enforce an anti-double-dipping rule, ensuring a student can only claim one meal per session.

VII. Evaluation Metrics

The success of these modifications will be evaluated iteratively during weekly lab sessions based on the EEE 312 rubric:

- **Accuracy Test:** The system must show a significantly lower False Rejection Rate (FRR) than the senior codebase when subjects vary their distance from the microphone.
- **Spoofing Test:** The system must successfully reject playback attacks using the Sub-200 Hz energy threshold check.
- **Technical Demonstration:** Live execution of the MATLAB script demonstrating the anti-double-dipping session logic and CSV logging.
- **Presentation & Report:** A final video presentation and report detailing the DSP methodology, mathematical logic, and code efficiency improvements over the baseline.

References

- [1] A. V. Oppenheim, R. W. Schaffer, and J. R. Buck, *Discrete-Time Signal Processing*, 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2009. (Concepts utilized: Z-Transform properties, LTI System Analysis, DFT computation).
- [2] J. G. Proakis and D. G. Manolakis, *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th ed. Upper Saddle River, NJ: Pearson Prentice Hall, 2006. (Concepts utilized: Sampling theorem, Quantization, Frequency Domain filtering, Spectral Subtraction).